Cyclic symmetry and failure of block-permutation invariance in Dexter intervals

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Abstract

Over every field, cyclic rotation of the blocks of a Dyck word preserves the bounded derived category of its Dexter interval $J(w)$. Arbitrary block permutation does not: an exact computation gives two 288-element intervals with distinct integral Euler polynomials. For the three blocks 110100, 11011000, and 1100, their six permutations consequently give exactly two derived equivalence classes, namely the two cyclic orbits. Boundary attachment explains the distinction. The lower truncations L_{uv} and L_{vu} for $u = 110100$, $v = 11011000$ are derived equivalent, but no enhanced equivalence between them intertwines their boundary pullbacks, even after an arbitrary enhanced autoequivalence of the base. We prove this using a relative extension criterion and the counterexample. Exchanges involving the elementary block and transfers between pyramid blocks provide positive supplementary families.

1 Introduction

A Dyck word is a word in $1, 0$ with nonnegative prefix heights and total height zero. Its blocks are its nonempty factors between consecutive returns to height zero. Write $b(x)$ for the number of blocks of x. We use the Dexter order of [1]. If a primitive Dyck subpath s is preceded by a zero and is followed by a one or by the end of the word, it is movable. A cover slides s to the left across any positive number of consecutive immediately preceding zeros. Such a subpath may be based at positive height; it need not be a whole block.

Let $I(w) = [w_{\min}, w]$ and

$$J(w) = [w_{\min}, (1, w, 1, 0, 0)].$$

For a finite poset P, $\mathrm{Rep}_k P$ denotes finite-dimensional covariant representations, and all equivalences are k-linear. The following assertion is the block-permutation subproblem explicitly proposed in [1, Section 9, Conjecture 9.1].

Problem 1.1 (Block-permutation assertion, disproved below)**.** *For every field k, every pair of nonempty indecomposable Dyck blocks u, v, and possibly empty block strings A, B, does one have*

$$D^b(\mathrm{Rep}_k J(AuvB)) \simeq D^b(\mathrm{Rep}_k J(AvuB))?$$

The positive symmetry is cyclic rotation: for all nonempty Dyck words u, v and every field k, Theorem 3.2 gives

$$D^b(\mathrm{Rep}_k J(uv)) \simeq D^b(\mathrm{Rep}_k J(vu)).$$

In contrast, with $u = 110100$, $v = 11011000$, and $b = 1100$, Theorem 4.2 separates $J(uvb)$ from $J(vub)$ by their integral Euler polynomials. Both intervals have 288 points. Corollary 4.4 combines

the two results into exactly two equivalence classes for one specified three-block multiset. No general classification by cyclic order is claimed.

The distinction is visible in the boundary data. Theorem 5.1 identifies the abstract derived categories of the lower truncations. The relative extension criterion, Proposition 5.3, then shows that for the counterexample pair their boundary embeddings cannot be intertwined by any enhanced equivalence and any enhanced base autoequivalence (Corollary 5.5). Thus the boundary-decorated category contains information absent from its abstract derived type. Elementary exchanges and pyramid transfers are supplementary positive results, not steps toward the disproved universal assertion.

The product decomposition of $I(w)$, the chain decomposition, and the canonical shape cut are inputs from [1, Proposition 4.4 and Section 5.2]. For general finite posets, Ladkani proved that the ordinal sums $X \oplus Y$ and $Y \oplus X$ are derived equivalent [3, Corollary 4.15]; associativity then gives cyclic rotation of any ordinal sum. His Example 4.20 shows that a permutation of three summands can fail to preserve derived equivalence. Thus cyclic symmetry together with permutation failure has a precedent for ordinal sums. The singleton flip-flop mechanism used in Appendix C is also known [2, Corollary 1.3], and the triangular-algebra tilting method has the background of [4, Theorems 4.5 and 4.9].

Here the objects are the specific Dexter intervals $J(w)$, whose mixed relations depend on their projecting boundary coordinates. The calculations below identify the cross-chain rules, the target mixed bimodules and their multiplication, and the natural boundary comparisons needed for rotation and pyramid transfer. The explicit block-permutation counterexample concerns these intervals. The ordinal-sum results alone do not identify the required Dexter objects and boundary data; their relation is not assumed in our proofs. Appendix calculations use different examples to explain limitations of particular cuts and restricted base corrections.

The paper follows this distinction. Section 2 gives the chain coordinates and the marked-boundary algebra. Section 3 proves cyclic symmetry; Section 4 then supplies the counterexample and the two cyclic orbits. Section 5 proves the abstract lower-truncation equivalence, the relative extension criterion, and their incompatible boundary consequence, in that order. Section 6 records the positive supplementary families. Computational details and the longer mutation argument are placed in the appendices.

2 Chain coordinates and the marked-boundary algebra

2.1 The fibres and their cross-chain arrows

For $x = x_1 \cdots x_b$ in blocks, put

$$\begin{aligned} e_i(x) &= (1, x_1, \dots, x_i, 0, x_{i+1}, \dots, x_b, 1, 0) && (0 \le i \le b), \\ e_{b+1}(x) &= (1, x, 1, 0, 0). \end{aligned}$$

We call the first $b+1$ points lower-shape points and the last point a top-shape point. This terminology avoids confusing shape labels with the prefix and suffix A, B.

Lemma 2.1. *The elements of $J(w)$ are precisely $e_i(x)$ with $x \le w$ and $0 \le i \le b(x)+1$. Each such chain is complete, and the map $q(e_i(x)) = x$ is order preserving. The top-shape subposet is $I(w)$.*

Proof. The shape subposet of [1, Lemmas 5.3–5.4] is a lower ideal partitioned by these chains. A cross-chain cover projects to a lower cover by Proposition 5.5 of that paper. Thus q is order

preserving. A point below the top $e_{b(w)+1}(w)$ must project below w. Conversely the top-shape embedding preserves order, so the entire chain above $x \le w$ lies below that top. □

A lower cover $a : x \prec y$ has one of two types. An internal cover slides a subpath based at positive height and leaves the block count unchanged. A merging cover slides a whole block x_ℓ, with $2 \le \ell \le b(x)$, into its preceding block. Then $b(y) = b(x) - 1$; we call ℓ the merging position. These exhaust the covers: a slide cannot cross a one, so a zero-height slide joins exactly two adjacent blocks.

Proposition 2.2 (Cross-chain cover rule)**.** *All covers of $J(w)$ are the following, with $x, y \le w$.*

1. *The vertical covers $e_i(x) \prec e_{i+1}(x)$.*
2. *For an internal cover $x \prec y$, the covers $e_i(x) \prec e_i(y)$ for $0 \le i \le b(x) + 1$.*
3. *For a merging cover $x \prec y$ at position ℓ, the covers*

$$e_i(x) \prec e_i(y) \quad (0 \le i < \ell), \qquad e_i(x) \prec e_{i-1}(y) \quad (\ell < i \le b(x) + 1).$$

There is no cross-chain cover over this lower cover with source $e_\ell(x)$.

Proof. Write a lower-shape point as $(1, p, 0, q, 1, 0)$, where $x = pq$ and p consists of the first i blocks. Call the zero between p and q the inserted zero. An internal movable subpath has positive baseline in x. It cannot end at a zero-height cut, and its preceding zero run cannot cross such a cut. Inserting the wrapper therefore preserves its movability and its slide, on either side of the cut. The index stays i.

Now slide the whole block x_ℓ across t zeros in x. If $i < \ell - 1$, the move occurs in q and the cut index is unchanged. If $i = \ell - 1$, slide it across $t + 1$ zeros in the wrapped word: the extra zero is the inserted zero. The resulting wrapped prefix ends after the merged block, at index $\ell - 1$ of y. If $i > \ell$, the move lies in p, and x_ℓ is followed there by a one. It remains movable; merging reduces the prefix block count by one. If $i = \ell$, however, x_ℓ is followed by the inserted zero. It is not movable. This is precisely the excluded case. The top-shape case is the unchanged lower slide inside the wrapper, with top index reduced by one when a merge occurs.

For completeness, consider any slide that remains in the shape subposet. A slide of the first block of q across just the inserted zero is a vertical cover. Sliding it farther gives exactly the merging case just described. All remaining slides away from the terminal 10 project to an internal or merging lower cover. A slide of that terminal 10 stays in the shape subposet only when q is empty and it crosses just the inserted zero; this is the last vertical cover. If q is nonempty, such a slide either leaves an earlier return to height zero or produces too many terminal zeros, and is neither of the two shapes. A longer terminal slide when q is empty also has too many terminal zeros. These observations exhaust the movable subpaths. Since the shape subposet is a lower ideal, its induced covers are precisely the ambient Dexter covers with both endpoints in it. Restricting to $J(w)$ introduces no new covers. The stated list is therefore complete. □

The excluded position cannot be repaired by pretending that every cut simply follows its image in the merged word. The correct formula for comparability allows vertical motion first. For each lower cover $a : x \prec y$ define a monotone map between index chains by

$$f_a(i) = \begin{cases} i, & a \text{ internal}, \\ i - \mathbf{1}_{i>\ell}, & a \text{ a merge at } \ell. \end{cases}$$

The domains include the top index $b(x) + 1$.

Corollary 2.3 (Full comparability)**.** *For $x \le y$ let*

$$m_{x,y}(i) = \min_{\pi : x \rightsquigarrow y} f_\pi(i), \qquad f_\pi = f_{a_r} \circ \cdots \circ f_{a_1},$$

where π runs through saturated lower paths; for $x = y$ use the empty path. Then

$$e_i(x) \le e_j(y) \quad \Longleftrightarrow \quad x \le y \text{ and } j \ge m_{x,y}(i).$$

Proof. Over one lower cover, $f_a(i)$ is the smallest reachable target index. The cover rule proves this directly except at $i = \ell$. There one first moves vertically to $\ell + 1$ and then crosses to index ℓ; if $\ell = b(x)$ the intermediate vertex is the top. No smaller target is reachable by the listed covers. Monotonicity shows that waiting longer on a fibre cannot improve this minimum. Any upper path projects, after deleting vertical steps, to a lower saturated path. Its final index is at least the corresponding composite $f_\pi(i)$. Conversely each such composite can be attained by lifting its covers in order; any larger final index is obtained by vertical motion. There are finitely many paths, so the minimum is attained. □

Remark 2.4. This formula takes a minimum over paths. It does not assume that different lower paths give the same map of chains, and does not assert a Grothendieck construction with path-independent transition functors.

2.2 Incidence algebras and exterior factors

Fix the category-algebra convention in which a basis element a_{ts} represents $s \le t$, and $a_{ut}a_{ts} = a_{us}$. Covariant representations are left modules. By Corollary 2.3, the algebra has basis a_{ts} for exactly the comparable chain points, and this composition law. Equivalently it is the path algebra of the quiver in Proposition 2.2, modulo the differences of all parallel paths. Indeed each comparable pair has a path, these relations identify all paths with those endpoints, and relations with distinct endpoints do not interact. The quotient therefore has precisely the claimed basis, including the identity paths.

The lower product decomposition gives

$$I(AuvB) = I(A) \times I(u) \times I(v) \times I(B),$$

where the empty word contributes a one-point factor. This is an actual product of posets, not a claim about $J(AuvB)$.

Corollary 2.5 (Context offsets)**.** *Suppose a cover occurs in one factor of this product, and let c be the total number of blocks in the current preceding factors. An internal move uses the identity index rule. A merging move at local position h uses the rule of Proposition 2.2 with global merging position $\ell = c + h$. The incidence-algebra presentation above includes all arrows and mixed compositions for arbitrary A, B.*

Proof. The block boundaries of the upper word must remain returns to zero throughout its lower interval. Thus a cover changes exactly one factor. Counting the blocks before that factor gives $\ell = c + h$; the complete cover rule and the parallel-path presentation apply. □

The offset c depends on the lower element, not only on the fixed upper word. Swapping product factors does not by itself transport these cover rules. Corollary 2.5 supplies the original attachments. Transport through a derived equivalence additionally requires the boundary comparison in Section 5.

2.3 Derived conventions and tilting

For a vertex x of a finite poset, write $P_x = k\,\mathrm{Hom}(x, -)$. Thus $\mathrm{Hom}(P_x, P_y) = k$ exactly when $y \le x$. For a tilting object indexed by a target poset, the algebra to identify is its opposite endomorphism algebra. We use the following specified finite-algebra functors. For a finite poset X, write $\mathcal{D}_X = D^b(\mathrm{Rep}_k X)$ and denote its Serre functor by $\mathbb{S}_X$. It is derived tensor with the k-dual of kX; finite global dimension makes it an autoequivalence. Derived Kan extensions along $q : X \to S$ are denoted

$$\ell_X \dashv Q_X \dashv r_X, \qquad Q_X = q^*.$$

They can be computed by finite poset bar resolutions. Their Serre adjunction identity is

$$r_X \mathbb{S}_X \simeq \mathbb{S}_S \ell_X. \tag{1}$$

Indeed Serre duality applied to $\mathrm{Hom}_X(Q_X V, \mathbb{S}_X E)$ identifies it with $\mathrm{Hom}_S(V, \mathbb{S}_S \ell_X E)$, naturally in both variables. In particular $\ell_X(P_a) = P_{q(a)}$. External tensor products and derived Hom below are over the field k. We use the standard differential graded enhancements by perfect module complexes. Enhanced equivalences and functor comparisons mean equivalences and isomorphisms of the corresponding derived bimodules; in particular they admit external tensor products and transport both actions of a gluing bimodule. All functors constructed below are represented in this framework. The negative boundary statement quantifies over this entire class of enhanced equivalences.

Lemma 2.6 (Tilting convention)**.** *Let T generate $\mathcal{D}_X$, satisfy $\mathrm{Hom}(T, T[m]) = 0$ for $m \neq 0$, and have $\mathrm{End}(T)^{\mathrm{op}} \cong kY$ for a finite poset Y, respecting the specified vertex idempotents. Then there is an enhanced equivalence $\mathcal{D}_Y \to \mathcal{D}_X$ sending the vertex projectives to the specified summands of T.*

Proof. A projective-complex representative of T gives a differential graded endomorphism algebra with cohomology only in degree zero. Its truncation provides a quasi-isomorphism zigzag to $\mathrm{End}(T)$. The Hom functor is an equivalence on the thick category generated by T: it is so on the generator and hence on finite cones and summands, whose image is the perfect module category. Generation and finite global dimension of the identified target poset algebra give the claimed bounded equivalence. □

2.4 Cutting at a word boundary

For a nonempty Dyck word z, put $S_z = I(z)$ and define two full subposets of $J(z)$:

$$L_z = \{e_i(x) : x \le z,\ 0 \le i \le b(x)\}, \qquad R_z = \{e_i(x) : x \le z,\ 1 \le i \le b(x) + 1\}.$$

Thus L_z removes the top-shape layer, whereas R_z removes the bottom layer of every chain. Both carry the restriction q_z of q. A poset subscript on a derived functor specifies its actual finite representation category.

Proposition 2.7 (The marked-boundary algebra)**.** *For nonempty Dyck words u, v, the poset $J(uv)$ is a two-piece poset with lower ideal $D = L_u \times S_v$ and upper filter $U = S_u \times R_v$. Its mixed relations are exactly*

$$(d, y) \le (x, r) \quad \Longleftrightarrow \quad q_u(d) \le x \text{ and } y \le q_v(r). \tag{2}$$

These are relations of the actual poset, including their incidence compositions.

Proof. Write a lower element as xy, with $x \le u$, $y \le v$, and put $a = b(x)$. Partition its chain at $i = a$: the point $e_i(xy)$ belongs to D if $i \le a$, with coordinates $(e_i(x), y)$; otherwise it belongs to U, with coordinates $(x, e_{i-a}(y))$. For a lower cover inside u, Proposition 2.2 restricts on D to the L_u rule and acts on U without changing the R_v index: both i and a decrease by one on a merging move. For a lower cover inside v, it restricts on U to the R_v rule and leaves the L_u index unchanged on D. A crossing can leave D for U, but none can go from U to D. This proves the claimed induced subposets and ideal/filter properties.

The section $h(x, y) = e_{b(x)+1}(xy)$ is monotone: merges inside u reduce its index by one, and merges inside v preserve it. It is the first point of U in its fibre. Every d lies below the section in its own fibre, and $h(x, y)$ lies below every upper-fibre point over $(x', y') \ge (x, y)$. Monotonicity of q proves the converse implication. This gives (2). The composition statement follows from the incidence basis, not from an independent choice of mixed scalars. □

3 Cyclic rotation

3.1 The two truncations and their relative equivalence

Lemma 3.1 (An explicit truncation equivalence)**.** *There is an equivalence $F_z : \mathcal{D}_{L_z} \to \mathcal{D}_{R_z}$ with a natural isomorphism*

$$F_z Q_{L_z} \simeq Q_{R_z} \mathbb{S}_{S_z}. \tag{3}$$

Consequently

$$\ell_{L_z} F_z^{-1} \simeq \mathbb{S}_{S_z}^{-1} \ell_{R_z}, \tag{4}$$

$$r_{R_z} F_z \mathbb{S}_{L_z} \simeq \mathbb{S}_{S_z}^{2} \ell_{L_z}. \tag{5}$$

All these identifications are identities of functors, with their maps, not only identities on object classes.

Proof. Work first in $\mathcal{C} = \mathcal{D}_{J(z)}$. Let $B = Q_{J(z)} : \mathcal{D}_{S_z} \to \mathcal{C}$ and let $Z : \mathcal{D}_{S_z} \to \mathcal{C}$ be extension from the top layer. For projectives they are respectively

$$B(P_x) = P_{e_0(x)}, \qquad Z(P_x) = P_{e_{b(x)+1}(x)}.$$

Thus both are fully faithful. The first formula follows because $e_0(x) \le e_i(y)$ exactly when $x \le y$; the second is extension from an upper filter. They have both adjoints on these bounded categories. Injectives at top points give the natural identity

$$\mathbb{S}_{J(z)} Z \simeq B \mathbb{S}_{S_z}.$$

Indeed the injective at the top over x has value k exactly on points projecting below x, with the usual incidence maps.

Use extension by zero to regard $\mathcal{D}_{L_z}$ and $\mathcal{D}_{R_z}$ as full derived subcategories of $\mathcal{C}$. These embeddings are fully faithful: the sets are respectively a lower ideal and an upper filter, and restriction of their projective resolutions verifies derived full faithfulness. They identify with $Z^{\perp}$ and $B^{\perp}$, respectively, by evaluation on the top and bottom projectives. Serre duality and the displayed identity give $Z^{\perp} = {}^{\perp}B$. Here orthogonality includes all shifts.

For $E \in {}^{\perp}B$, form the actual counit cone

$$H_z(E) = \mathrm{Cone}(BB^R E \longrightarrow E).$$

It lies in $B^{\perp}$. This defines an equivalence ${}^{\perp}B \to B^{\perp}$. To check full faithfulness, apply Hom to the two counit triangles: $\mathrm{Hom}(E, B(-)[m]) = 0$ and $\mathrm{Hom}(B(-), H_z(E')[m]) = 0$ identify the Hom groups with those before mutation. An inverse on $G \in B^{\perp}$ is the fibre of the unit $G \to BB^{L}G$; it belongs to ${}^{\perp}B$, and the same triangles show that the two operations are inverse. These cones and fibres are taken in the derived module enhancement.

The exact sequence $0 \to Z(P_x) \to B(P_x) \to Q_{L_z}(P_x) \to 0$ shows, naturally on the projective generators of $\mathcal{D}_{S_z}$, that

$$H_z Q_{L_z} \simeq Z_{R_z}[1],$$

where Z_{R_z} is the top-layer projective embedding into R_z. In R_z the same injective calculation gives $\mathbb{S}_{R_z} Z_{R_z} \simeq Q_{R_z} \mathbb{S}_{S_z}$. Notice that $Q_{R_z}(P_x) = P_{e_1(x)}$; the index 1 is preserved by every lower cover. Set

$$F_z = \mathbb{S}_{R_z} H_z[-1].$$

This proves (3). Taking left adjoints gives (4). Taking right adjoints first gives $r_{R_z} F_z \simeq \mathbb{S}_{S_z} r_{L_z}$; composing with $\mathbb{S}_{L_z}$ and applying (1) gives (5). Naturality follows from the actual counit, unit and Serre pairings used throughout. □

3.2 The rotation tilting object

The next result treats two factors without exterior strings. The factors may themselves have several blocks, so it also proves cyclic rotation of an arbitrary block string. Its two-piece tilting construction uses the triangular-algebra method of [4, Theorems 4.5 and 4.9] as background. We compute the actual mixed bimodule and identify the target $kJ(vu)$, including multiplication, in the proof below.

Theorem 3.2 (Two-factor rotation)**.** *For all nonempty Dyck words u, v and all fields k there is an equivalence*

$$D^b(\mathrm{Rep}_k J(uv)) \simeq D^b(\mathrm{Rep}_k J(vu)).$$

Proof. Use Proposition 2.7, with $D = L_u \times S_v$ and $U = S_u \times R_v$. Let $j_0 : \mathcal{D}_D \to \mathcal{D}_{J(uv)}$ and $i : \mathcal{D}_U \to \mathcal{D}_{J(uv)}$ be extension by zero. The first is the lower-supported embedding and the second the upper-supported embedding. For $E \in \mathcal{D}_D$ and $V \in \mathcal{D}_U$, the triangular algebra gives

$$R\,\mathrm{Hom}(iV, j_0 E) = 0, \tag{6}$$

$$R\,\mathrm{Hom}(j_0 E[-1], iV) \simeq R\,\mathrm{Hom}_U(\Phi E, V), \qquad \Phi = \ell_{L_u} \boxtimes Q_{R_v}. \tag{7}$$

Here is a direct justification of the second formula. Derived extension of the lower projectives gives a triangle $i\Phi E \to j_! E \to j_0 E$; its upper restriction is the actual mixed bimodule of (2). On $P_d \boxtimes P_y$ that bimodule gives $P_{q_u(d)} \boxtimes Q_{R_v} P_y$, which proves the formula for Φ, including its actions, by projective resolutions. Adjunction gives $R\,\mathrm{Hom}(j_! E, iV) = 0$, and hence (7). The zero in (6) follows because upper projectives stay projective in the full poset and have no maps to a lower-supported object.

Index the desired summands by the marked-boundary decomposition of the target $J(vu)$: its lower vertices are $(d, x) \in L_v \times S_u$, and its upper vertices are $(y, r) \in S_v \times R_u$. Define in the source category

$$T_{d,x} = i\big(\mathbb{S}_{S_u}^{-1} P_x \boxtimes F_v \mathbb{S}_{L_v} P_d\big),$$
$$T_{y,r} = j_0\big(F_u^{-1} P_r \boxtimes \mathbb{S}_{S_v}^{2} P_y\big)[-1].$$

This is a uniform construction by finite derived tensor products, Kan extensions and the counit cones of Lemma 3.1. It is not specified by assuming the required endomorphism algebra. Each factor is a bounded complex over a finite poset algebra.

Inside each family, derived full faithfulness and the equivalences F_z and $\mathbb{S}_X$ identify all Hom complexes with those of the corresponding target projectives. The Hom complexes from the first family to the second vanish by (6). For the remaining direction, (7), the tensor-Hom identity over k, and (4)–(5) give

$$\begin{aligned} R\operatorname{Hom}(T_{y,r}, T_{d,x}) &\simeq R\operatorname{Hom}_{S_u}(\ell_{L_u} F_u^{-1} P_r, \mathbb{S}_{S_u}^{-1} P_x) \\ &\quad \otimes_k R\operatorname{Hom}_{S_v}(\mathbb{S}_{S_v}^2 P_y, r_{R_v} F_v \mathbb{S}_{L_v} P_d) \\ &\simeq \operatorname{Hom}_{S_u}(P_{q_u(r)}, P_x) \otimes_k \operatorname{Hom}_{S_v}(P_y, P_{q_v(d)}). \end{aligned}$$

It is concentrated in degree zero, of dimension one precisely when $x \le q_u(r)$ and $q_v(d) \le y$. These are exactly the mixed relations of the target, in the projective convention.

The displayed comparison is natural in all four variables and in the projective maps. It is therefore an isomorphism of the mixed bimodule with the target incidence bimodule, not just an equality of dimensions. Choose its basis by the tensor of the two canonical incidence maps. Composition with maps in either family agrees, by this naturality, with incidence multiplication. Products inside each family were already identified by the equivalences; there are no products of two mixed arrows. Thus, for the sum T of all the displayed objects, we have the actual algebra isomorphism

$$\operatorname{End}(T)^{\mathrm{op}} \cong kJ(vu).$$

The preceding calculations prove $\operatorname{Hom}(T, T[m]) = 0$ for $m \neq 0$.

Each family generates its entire supported category, because it is the image of projective generators under equivalences and external tensor products. These two supported categories generate the full category: every representation has its upper-supported subrepresentation and lower-supported quotient, and the same statement follows for bounded complexes by truncation. Hence T is tilting. Lemma 2.6 gives the enhanced equivalence. All constructions use finite-dimensional algebras over k, with no characteristic-dependent averaging or division. This proves the theorem for every field. □

Corollary 3.3. *Cyclic rotations of the block string of w preserve $D^b(\operatorname{Rep}_k J(w))$ over every field. In particular Problem 1.1 holds when A and B are empty.*

Proof. Cut the string into the two words before and after the desired rotation and apply Theorem 3.2. □

4 Failure of block-permutation invariance

The obstruction in this section concerns the complete intervals $J(w)$, not a relative boundary condition or a chosen tilting construction. All calculations use integers; the auxiliary prime is unrelated to the characteristic of the representation field.

Lemma 4.1 (Integral Euler obstruction)**.** *Let $C_P = (1_{x \le y})$ be the zeta matrix of a finite poset P in a linear extension. The polynomial*

$$F_P(t) = \det(tC_P^t - C_P) \in \mathbb{Z}[t]$$

is invariant under k-linear derived equivalence of its incidence algebra, for every field k.

Proof. The simple classes give an integral basis of K_0, and the Euler matrix in this basis is $E_P = C_P^{-1}$. Indeed, covariant projectives satisfy $\mathrm{Hom}(P_x, S_y) = k\delta_{xy}$ and have dimension vectors given by the rows of C_P. Incidence algebras have finite global dimension, so the Euler form is defined on the bounded derived category. A k-linear equivalence induces an integral unimodular change of basis preserving this form. Consequently $\det(tE_P - E_P^t)$ is invariant. Finally,

$$C_P^t(tE_P - E_P^t)C_P = tC_P^t - C_P, \qquad \det C_P = 1.$$

This proves the claim independently of any convention for the sign of the Coxeter transformation. □

Theorem 4.2 (Computational counterexample)**.** *Set*

$$u = 110100, \quad v = 11011000, \quad A = \varnothing, \quad B = 1100.$$

These are admissible blocks and exterior strings in Problem 1.1. For every field k there is no k-linear triangulated equivalence

$$D^b(\mathrm{Rep}_k\, J(uvB)) \simeq D^b(\mathrm{Rep}_k\, J(vuB)).$$

In particular the block-permutation assertion of [1, Section 9, Conjecture 9.1] is false.

Proof by exact finite computation. The prefix heights of u and v are positive except at their final position, so both are indecomposable. The complete Dexter intervals below $1uvB100$ and $1vuB100$ each have 288 vertices and 882 covers. Their zeta matrices give the following exact residues:

w	$\lvert J(w)\rvert$	covers	$F_{J(w)}(2) \bmod 1009$
110100110110001100	288	882	238
110110001101001100	288	882	676

The different residues imply different integer polynomials, and Lemma 4.1 rules out equivalence over every field.

Appendix A explains the algorithm and checks; the self-contained implementation and full interval certificates are provided as supplementary files accompanying the manuscript. The calculation uses the direct Dexter move rule and does not depend on any positive equivalence in this paper. □

Remark 4.3. No minimality claim is made. The counterexample is outside both supplementary positive ranges: it has three non-elementary blocks, of which two are non-pyramid. The weaker successes for rotation and special block exchanges therefore do not imply the original universal assertion.

Corollary 4.4 (Two cyclic orbits, two derived types)**.** *Let $u = 110100$, $v = 11011000$, and $b = 1100$. For every field k, the six intervals obtained by permuting these blocks form exactly two k-linear derived equivalence classes:*

$$\{J(uvb), J(vbu), J(buv)\}, \qquad \{J(vub), J(ubv), J(bvu)\}.$$

Proof. Corollary 3.3 identifies the three members of each set. Theorem 4.2 separates the displayed representatives $J(uvb)$ and $J(vub)$. Since the three blocks are distinct, the two sets exhaust all six permutations. No further computations are needed. □

For this multiset the derived category detects the two cyclic orders. This statement concerns these six words only.

5 Abstract equivalence and boundary attachment

The lower truncations can be equivalent without admitting any boundary-compatible equivalence. We first construct the abstract equivalence, then show what boundary compatibility would imply, and finally use the counterexample to rule it out.

5.1 Rotating the lower truncation

Put

$$W_z = \{e_i(x) : x \le z,\ 1 \le i \le b(x)\} \subset L_z.$$

This removes the bottom layer as well as the top-shape layer. The pullback Q_{W_z} sends P_x to $P_{e_1(x)}$, so is fully faithful. Let $s_z : \mathcal{D}_{S_z} \to \mathcal{D}_{L_z}$ be extension by zero to the bottom layer. The bottom layer is a lower ideal: a chain index cannot decrease from a positive value to zero along a cover. Since $Q_{L_z} P_x = P_{e_0(x)}$, Serre sends it to the injective supported on bottom vertices over $y \le x$. This identifies its maps as well as its support, and gives

$$\mathbb{S}_{L_z} Q_{L_z} \simeq s_z \mathbb{S}_{S_z}. \tag{8}$$

In particular s_z has a right adjoint s_z^R, and

$$s_z^R \mathbb{S}_{L_z}^2 \simeq \mathbb{S}_{S_z} r_{L_z} \mathbb{S}_{L_z} \simeq \mathbb{S}_{S_z}^2 \ell_{L_z}. \tag{9}$$

The base Serre functor in these identities is retained in the mixed Hom calculation below.

Theorem 5.1 (Lower-truncation rotation)**.** *For nonempty Dyck words u, v there is an enhanced equivalence*

$$\mathcal{T}_{u,v} : \mathcal{D}_{L_{vu}} \longrightarrow \mathcal{D}_{L_{uv}}.$$

It has the explicit boundary value

$$\mathcal{T}_{u,v} Q_{L_{vu}} \simeq \mathcal{E}_{u,v}(\mathrm{id}_{\mathcal{D}_{S_u}} \boxtimes \mathbb{S}_{L_v}^2 Q_{L_v}), \tag{10}$$

where the base factors on the left are identified with $S_u \times S_v$, and $\mathcal{E}_{u,v}$ is the padding functor constructed below. Formula (10) is not asserted to be a constant pullback for general u, v.

Proof. The lower truncation has three consecutive layers

$$S_u \times S_v, \qquad W_u \times S_v, \qquad S_u \times W_v.$$

Indeed, over (x, y) with $a = b(x)$, these correspond respectively to $i = 0$, $1 \le i \le a$, and $a < i \le a + b(y)$. The induced orders and mixed relations follow by restricting Proposition 2.7; between two different layers the relation is precisely comparison of the two base coordinates. The layer unions in this order are lower ideals.

Define a monotone map

$$\rho : L_{uv} \longrightarrow S_u \times L_v, \qquad e_i(xy) \longmapsto \begin{cases} (x, e_0(y)), & i \le b(x), \\ (x, e_{i-b(x)}(y)), & i > b(x). \end{cases}$$

The cover rule proves monotonicity. Let $\mathcal{E}_{u,v} = \rho^*$, which is exact. It sends $P_x \boxtimes P_{e_0(y)}$ to the bottom projective of L_{uv} over (x, y), and sends $P_x \boxtimes P_b$, $b \in W_v$, to the projective in its last layer. These are actual representable maps, so $\mathcal{E}_{u,v}$ is derived fully faithful. It retains both the first and third layers, filling the middle layer by pullback of the first-layer coefficients.

Let $a_0 : \mathcal{D}_{W_u \times S_v} \to \mathcal{D}_{L_{uv}}$ be extension by zero to the middle layer. It is derived fully faithful. More explicitly, for a middle projective $P_a \boxtimes P_y$ there is an exact sequence

$$0 \longrightarrow i_B(P_{q_u(a)} \boxtimes Q_{W_v} P_y) \longrightarrow P^{L_{uv}}_{(a,y)} \longrightarrow a_0(P_a \boxtimes P_y) \longrightarrow 0, \tag{11}$$

where i_B is extension from the last layer. Both terms on the left are projective in the full poset. Their Hom into another middle-supported object computes exactly the middle Hom; this also proves the full-faithfulness assertion.

There are no derived maps from the image of $\mathcal{E}_{u,v}$ to the image of a_0: the displayed projective generators of the first image evaluate to zero on a middle-supported object. The reverse Hom has a useful precise form:

$$R\operatorname{Hom}\big(a_0(P_a \boxtimes Y), \mathcal{E}_{u,v}(P_x \boxtimes Z)\big) \simeq \operatorname{Hom}_{S_u}(P_{q_u(a)}, P_x) \otimes_k R\operatorname{Hom}_{L_v}(s_v Y, Z). \tag{12}$$

To verify this with its maps, first take $Y = P_y$. Applying Hom to (11) gives the fibre of

$$\operatorname{Hom}_{S_u}(P_{q_u(a)}, P_x) \otimes Z(e_0(y)) \longrightarrow \operatorname{Hom}_{S_u}(P_{q_u(a)}, P_x) \otimes R\operatorname{Hom}_{W_v}(Q_{W_v} P_y, Z|_{W_v}).$$

The map is the actual attachment of Z from its bottom layer. The exact sequence $0 \to i_{W_v} Q_{W_v} P_y \to Q_{L_v} P_y \to s_v P_y \to 0$ identifies this fibre with the right side of (12). This argument is natural in all projective maps and extends to bounded Y, Z by projective resolutions.

The target L_{vu} has a lower piece $L_v \times S_u$ and an upper piece $S_v \times W_u$. For their vertices define

$$T_{d,x} = \mathcal{E}_{u,v}(P_x \boxtimes \mathbb{S}^2_{L_v} P_d), \qquad T_{y,a} = a_0(P_a \boxtimes \mathbb{S}^2_{S_v} P_y).$$

Internal Hom complexes are those of the two target projective families. The Hom complexes in one mixed direction vanish as above. In the other direction, (12) and (9) identify $R\operatorname{Hom}(T_{y,a}, T_{d,x})$ with

$$\operatorname{Hom}_{S_u}(P_{q_u(a)}, P_x) \otimes_k \operatorname{Hom}_{S_v}(P_y, P_{q_v(d)}).$$

It is concentrated in degree zero, with exactly the target incidence condition. Naturality of the fibre calculation identifies both module actions and their compositions. Together with the internal algebra identifications, this proves $\operatorname{End}(T)^{\mathrm{op}} \cong kL_{vu}$, with no unverified scalar multiplication choices.

The two families generate the images of $\mathcal{E}_{u,v}$ and a_0. The first image contains the bottom and last-layer projectives, and (11) recovers all middle projectives. Thus the families generate the full source category and form a tilting object. The resulting equivalence sends target projectives to the indicated summands. Applying this to the target bottom projectives gives (10). More explicitly, a base projective $P_x \boxtimes P_y$ is sent to $\mathcal{E}_{u,v}(P_x \boxtimes \mathbb{S}^2_{L_v} P_{e_0(y)})$. For $x \le x'$ and $y \le y'$, the map from the primed projective to the unprimed one is the image under $\mathcal{E}_{u,v}$ of the representable map in the first factor tensored with $\mathbb{S}^2_{L_v}(P_{e_0(y')} \to P_{e_0(y)})$. These are exactly the internal maps of the first tilting family; the mixed bimodule identification above is natural for these maps on both sides. Thus no separate scalar choices are made on the base arrows. The identification is one of bimodules on the base projective category and therefore extends to the claimed enhanced functor identity by bounded projective resolutions. □

5.2 Extending a boundary-compatible equivalence

Proposition 5.2 (One-sided reduction)**.** *For fixed nonempty Dyck words u, v, suppose that $\mathcal{D}_{J(uvC)} \simeq \mathcal{D}_{J(vuC)}$ for every nonempty block string C. Then the adjacent exchange holds with all possibly empty exterior strings A, B.*

Proof. Put $C = BA$. Theorem 3.2, or the identity if A is empty, gives

$$\mathcal{D}_{J(AuvB)} \simeq \mathcal{D}_{J(uvBA)}, \qquad \mathcal{D}_{J(AvuB)} \simeq \mathcal{D}_{J(vuBA)}.$$

Apply the assumed suffix comparison when BA is nonempty. When BA is empty use Theorem 3.2 itself. No compatibility between these rotations is required for existence of the resulting equivalence. □

The required suffix bimodule has an exact description. More generally, let w, w' be nonempty Dyck words with a specified poset identification $S = I(w) \cong I(w')$. For an exchanged pair this is the product-factor permutation. All functors to and from S below use the specified identification. For a nonempty suffix C, put $V = I(C)$. By Proposition 2.7, the lower and upper pieces of $J(wC)$ are

$$D_w = L_w \times V, \qquad U = S \times R_C.$$

The upper piece is literally common to the two words after the factor identification. The actual mixed derived tensor functor is

$$\Phi_w = \ell_{L_w} \boxtimes Q_{R_C} : \mathcal{D}_{D_w} \longrightarrow \mathcal{D}_U. \tag{13}$$

Thus replacing L_w by a derived-equivalent category without controlling ℓ_{L_w} does not identify the gluing algebra. The following criterion transports this specified gluing bimodule. It uses triangular-algebra tilting, as in [4, Theorems 4.5 and 4.9], with the natural boundary comparison and the Dexter target identification checked here.

Proposition 5.3 (Relative extension criterion)**.** *Suppose an enhanced equivalence $G : \mathcal{D}_{L_w} \to \mathcal{D}_{L_{w'}}$ has been constructed, an enhanced autoequivalence K of $\mathcal{D}_S$, and a natural isomorphism of functors*

$$GQ_{L_w} \simeq Q_{L_{w'}} K : \mathcal{D}_S \longrightarrow \mathcal{D}_{L_{w'}}. \tag{14}$$

Then for every nonempty C there is a tilting object in $\mathcal{D}_{J(wC)}$ with opposite endomorphism algebra $kJ(w'C)$. The same comparison gives $\mathcal{D}_{J(AwB)} \simeq \mathcal{D}_{J(Aw'B)}$ for arbitrary exterior block strings A, B.

Proof. Taking left adjoints in (14) gives a natural isomorphism

$$\ell_{L_w} G^{-1} \simeq K^{-1} \ell_{L_{w'}}.$$

It identifies the mixed functors, with their two module actions:

$$\Phi_w (G^{-1} \boxtimes \mathrm{id}) \simeq (K^{-1} \boxtimes \mathrm{id}) \Phi_{w'}. \tag{15}$$

Let $j_!$ be derived extension from the lower piece D_w, and let i be extension by zero from the upper piece U in $J(wC)$. For target lower vertices $(d, y) \in L_{w'} \times V$ and target upper vertices $(x, r) \in S \times R_C$, define

$$T_{d,y} = j_!(G^{-1} P_d \boxtimes P_y), \qquad T_{x,r} = i(K^{-1} P_x \boxtimes P_r).$$

Here $j_!$ is essential: replacing it by lower-supported extension by zero would change the mixed Hom and its degree. The lower restriction of $j_! E$ is E, and its upper restriction is $\Phi_w E$. Hence $j_!$ and i are fully faithful, and

$$R\,\mathrm{Hom}(j_! E, iW) = 0, \qquad R\,\mathrm{Hom}(iW, j_! E) \simeq R\,\mathrm{Hom}_U(W, \Phi_w E).$$

Using (15), the remaining mixed calculation is

$$\begin{aligned} R\operatorname{Hom}(T_{x,r}, T_{d,y}) &\simeq R\operatorname{Hom}_U(K^{-1}P_x \boxtimes P_r, K^{-1}P_{q_{w'}(d)} \boxtimes Q_{R_C}P_y) \\ &\simeq \operatorname{Hom}_S(P_x, P_{q_{w'}(d)}) \otimes_k \operatorname{Hom}_{R_C}(P_r, Q_{R_C}P_y). \end{aligned}$$

It is concentrated in degree zero, with value k precisely when $q_{w'}(d) \le x$ and $y \le q_C(r)$. This is the target mixed incidence relation in the projective convention. Internal Hom complexes are those of the target projectives by full faithfulness. Naturality of (15) identifies composition with both internal algebras. There are no products of two mixed arrows, so these comparisons give $\operatorname{End}(T)^{\mathrm{op}} \cong kJ(w'C)$ including multiplication. All nonzero-degree Hom groups vanish.

The two families generate the images of $j_!$ and i. The triangle $i\Phi_w E \to j_!E \to j_0E$ then recovers every lower-supported object; lower- and upper-supported objects generate the full bounded category. Thus T is tilting. For the empty suffix, use the lower part L_w and the top layer S of $J(w)$: the same calculation has mixed functor ℓ_{L_w} and no suffix tensor factor. Finally rotate AwB to wBA, and $Aw'B$ to $w'BA$, by Theorem 3.2; omit this rotation when A is empty. The suffix result gives the final assertion. □

Remark 5.4 (Boundary data are essential)**.** The extension criterion is a sufficient theorem with an explicit mixed bimodule, not an assumption imposed to obtain a universal exchange. Corollary 5.5 shows that its hypothesis fails for every possible pair (G, K) in the counterexample. Equality on K_0, or separate isomorphisms on the values of GQ, cannot replace the natural comparison of both module actions in (14).

5.3 The obstruction for every base correction

Corollary 5.5 (Abstract equivalence does not preserve boundary type)**.** *Let $u = 110100$, $v = 11011000$, and identify $S = I(uv) \cong I(vu)$ by permutation of the two product factors. For every field k there is an enhanced equivalence $\mathcal{D}_{L_{uv}} \simeq \mathcal{D}_{L_{vu}}$. Nevertheless there is no pair consisting of an enhanced equivalence $G : \mathcal{D}_{L_{uv}} \to \mathcal{D}_{L_{vu}}$ and an enhanced autoequivalence $K : \mathcal{D}_S \to \mathcal{D}_S$ with a natural isomorphism*

$$GQ_{L_{uv}} \simeq Q_{L_{vu}}K.$$

Proof. The inverse of Theorem 5.1 gives the abstract equivalence. If the stated pair existed, Proposition 5.3 with suffix 1100 would give an equivalence between $J(uv1100)$ and $J(vu1100)$, contradicting Theorem 4.2. The quantifier on K includes all enhanced base autoequivalences, not merely products of Serre powers. The enhancement and naturality requirements are exactly those of the extension criterion. □

6 Positive supplementary families

6.1 Exchanges involving an elementary block

The boundary value in Theorem 5.1 can be corrected when one factor is the elementary block $t = 10$. This is a genuine exterior-string case, not only a rotation of the whole word.

Theorem 6.1 (An elementary block with arbitrary exterior strings)**.** *Let $t = 10$. For every nonempty Dyck word u, every pair of possibly empty block strings A, B, and every field k, there is an equivalence*

$$\mathcal{D}_{J(AutB)} \simeq \mathcal{D}_{J(AtuB)}.$$

Proof. The base S_t is a point and L_t is the two-point chain $0 < 1$. Its bottom projective $P_0 = Q_{L_t}k$ satisfies

$$\mathbb{S}^2_{L_t} P_0 \simeq P_1[1]. \tag{16}$$

For an explicit check, $\mathbb{S}P_1 = P_0$ and $\mathbb{S}P_0$ is the bottom simple; applying Serre to the projective resolution of that simple gives (16).

In L_{ut} the last layer has one point $e_{b(x)+1}(xt)$ over each $x \in S_u$. It is a monotone top section, and every point projecting below x lies below this section over x. Let $Z_u : \mathcal{D}_{S_u} \to \mathcal{D}_{L_{ut}}$ be extension from this top section. Its projectives are those of the last layer. Top injectives therefore give, with their actual maps,

$$\mathbb{S}_{L_{ut}} Z_u \simeq Q_{L_{ut}} \mathbb{S}_{S_u}. \tag{17}$$

There is no additional top-shape layer in this assertion: it is an identity inside the lower truncation L_{ut}.

Equations (10) and (16) identify

$$\mathcal{T}_{u,t} Q_{L_{tu}} \simeq Z_u[1].$$

Consequently the actual equivalence

$$H_u = \mathbb{S}_{L_{ut}} \mathcal{T}_{u,t}[-1] : \mathcal{D}_{L_{tu}} \longrightarrow \mathcal{D}_{L_{ut}}$$

satisfies $H_u Q_{L_{tu}} \simeq Q_{L_{ut}} \mathbb{S}_{S_u}$. Its inverse $G_u = H_u^{-1}$ satisfies the relative comparison

$$G_u Q_{L_{ut}} \simeq Q_{L_{tu}} \mathbb{S}^{-1}_{S_u}.$$

These are natural isomorphisms, obtained on base projectives from representable and injective maps and then extended through the specified derived functors. Thus Proposition 5.3 applies with $K = \mathbb{S}^{-1}_{S_u}$. It constructs the suffix tilting object and identifies its endomorphism algebra for every nonempty suffix. The empty-suffix case is Theorem 3.2. Proposition 5.2 now supplies all A, B. No step uses division by the characteristic of k. □

Corollary 6.2. *If a Dyck word has at most two blocks different from* 10*, all permutations of its blocks give derived-equivalent J intervals over every field.*

Proof. Use Theorem 6.1 to move every elementary block to the right. With zero or one other block this gives one normal form. With two other blocks u, v the two possible forms are uvt^m and vut^m. Rotate the first to $vt^m u$ using Theorem 3.2, and move the elementary blocks across u using Theorem 6.1. This gives the second form, including the case $m = 0$. □

Remark 6.3 (Scope)**.** The elementary construction supplies an actual pair (G, K). Corollary 5.5 proves that no such pair can exist uniformly for general blocks. The next result establishes the additional pyramid family without asserting any extension to all indecomposable blocks.

6.2 Pyramid transfers and exchanges

Write $p_m = 1^m 0^m$ for $m \geq 1$; in particular $p_1 = 10$.

Theorem 6.4 (Pyramid transfer)**.** *For $a \geq 1$, $b \geq 2$, every field k, and arbitrary exterior block strings A_0, B_0, there is an equivalence*

$$\mathcal{D}_{J(A_0 p_a p_b B_0)} \simeq \mathcal{D}_{J(A_0 p_{a+1} p_{b-1} B_0)}.$$

Proof. Appendix B.1 identifies the two actual lower truncations with the posets of Theorem B.1 and proves the required natural boundary comparison. Applying Proposition 5.3 then gives the assertion in every exterior context. □

Corollary 6.5. *Any two pyramid blocks can be exchanged in arbitrary exterior strings, over every field. Consequently all block permutations are covered for words with at most one non-pyramid block.*

Proof. Successive transfers identify both pairs (p_a, p_b) and (p_b, p_a) with (p_{a+b-1}, p_1) in the same exterior context; a pair whose second index is one already has that form. For the last assertion, use two-factor rotation to place the possible non-pyramid block first. The remaining pyramid blocks can be permuted by the proved adjacent exchanges, and rotation places the exceptional block at any desired position. □

Remark 6.6 (Scope of the new construction)**.** This closes an infinite family containing arbitrarily many non-elementary blocks and arbitrary exterior strings. It does not settle arbitrary pairs of indecomposable Dyck blocks: their lower intervals need not admit the Boolean marked-cut model used in the transfer proof. No such model is assumed for a general block. The unrestricted Problem 1.1 is false by Theorem 4.2; that example has two non-pyramid blocks and lies outside this construction.

A The exact computational certificate

This appendix specifies the finite calculation used in Theorem 4.2. It requires no chain-coordinate formula and no tilting construction. Generate all Dyck words of semilength 11 by recursively appending 1 when fewer than 11 have been used, and 0 when the current height is positive. Match each opening step to its closing step by a stack. For every matched substring s, if its preceding letter is 0 and its following letter is 1 or absent, move s left past each possible positive number of consecutive preceding zeros. These are precisely the Dexter covers in the definition. Store the resulting incoming edges, and take the reverse reachable set from each of the two specified upper words. This finite graph has 58786 vertices before restricting to either interval; the two restricted cover lists are part of the certificate.

An upward move strictly increases the binary word lexicographically. Thus sorting the selected vertices gives a linear extension. Starting with each vertex itself, accumulate its upward reachable set in reverse lexicographic order; this produces C. Evaluate $\det(2C^t - C)$ by integer fraction-free elimination. With previous pivot d (initially 1) and current pivot a_{kk}, each trailing entry is replaced by

$$a_{ij} \longleftarrow \frac{a_{kk}a_{ij} - a_{ik}a_{kj}}{d} \qquad (i, j > k).$$

Use a row interchange when necessary, record its sign, and take the last diagonal entry with that sign. Every division is checked exact. Reducing the resulting integers modulo 1009 gives the values in Theorem 4.2.

Supplementary files and reproduction

The following two files are supplied with the manuscript in the `supplementary/` directory:

- `verify_counterexample_independent.py`, the self-contained implementation of the direct move algorithm and integer elimination;

- `independent_counterexample_certificate.json`, containing both full vertex and cover lists, both integer determinants, and the published control results.

The accompanying `README.md` records the run instructions. The implementation was tested with Python 3.12.10 and uses only the standard-library modules `json` and `pathlib`. From the manuscript directory, run

```
python supplementary/verify_counterexample_independent.py
```

Equivalently, from the `supplementary/` directory, run

```
python verify_counterexample_independent.py
```

Run Python with assertions enabled, without the `-O` option. The script writes the certificate JSON beside itself and overwrites that file on each run, independently of the working directory. The expected output, in the order $(\text{vertices}, \text{covers}, F(2) \bmod 1009)$, is

w	expected tuple
110100110110001100	$(288, 882, 238)$
110110001101001100	$(288, 882, 676)$

The two control intervals each have 27 vertices and $F(2) = 95869805$. The implementation imports none of the chain-coordinate implementations used elsewhere in this paper.

Checks on the computation

The computed vertex and cover sets agree exactly with those from reverse moves and with Proposition 2.2. An independent modular elimination gives the same residues. As a published control, for $J(11011000)$ and $J(11101000)$ this direct implementation gives 27 vertices and $F(2) = 95869805$ in both cases, agreeing with the Coxeter polynomial $\Phi_2\Phi_4\Phi_{18}\Phi_{54}$ stated in [1, Section 9], after the change from Coxeter to Serre sign. The finite computation is an explicit part of this proof, rather than a proposed symbolic identification of the characteristic polynomials.

B The relative mutation underlying pyramid transfer

This construction changes a specified family of cross relations. Its input is a specified finite poset, not a presumed relative equivalence. It will apply to a transfer between two pyramid blocks.

Let R be a finite poset with a least element. Let A, B be finite posets with monotone maps q_A, q_B to R. Assume that A has a bottom copy $\alpha(R)$ and that B has a bottom copy of R, both lower ideals, with

$$\alpha(r) \le a \iff r \le q_A(a), \qquad r \le b \iff r \le q_B(b), \qquad q_A\alpha = q_B|_R = \mathrm{id}_R.$$

Put $V = B \setminus R$. Form

$$D = A^0 \sqcup \widehat{R} \sqcup B^0, \qquad U = A^1 \sqcup B^1.$$

Within D and U, retain the given internal orders and compare vertices from an earlier displayed piece to a later displayed piece by comparison of their images in R, with no relations in the reverse direction. Define P with lower ideal D and upper filter U by the following mixed relations, and no others:

$$a^0 \le (a')^1 \iff a \le a', \qquad a^0 \le b^1 \iff q_A(a) \le q_B(b),$$
$$\widehat{r} \le b^1 \iff r \le q_B(b), \qquad b^0 \le (b')^1 \iff b \le b' \text{ and } b' \in V.$$

Let P' have the same pieces and relations except that its last condition is $b \le b'$, without the requirement $b' \in V$. These relations are transitive: internal comparisons use monotonicity of q_A, q_B, and the restricted last mixed relation is stable under postcomposition because V is an upper filter in B. Precomposition from A^0 or $\widehat{R}$ is covered by the other displayed mixed relations. No relation points backwards across the ordered pieces, so antisymmetry also holds. Thus both constructions are posets. Projection gives monotone maps $P, P' \to R \times C_2$, where $C_2 = \{0 < 1\}$ records D or U. Write $Q_P, Q_{P'}$ for their pullbacks.

Theorem B.1 (Relative two-term mutation)**.** *There is an enhanced equivalence $G : \mathcal{D}_P \to \mathcal{D}_{P'}$ with a natural identification*

$$GQ_P \simeq Q_{P'}(\mathrm{id}_{\mathcal{D}_R} \boxtimes \mathbb{S}_{C_2})[-1]. \tag{18}$$

This construction works over every field.

Proof. All projectives in the construction below belong to P. For $d \in A^0 \sqcup \widehat{R} \sqcup V^0$ define

$$f(a^0) = a^1, \qquad f(\widehat{r}) = r^1, \qquad f(v^0) = v^1, \qquad C_d = \mathrm{Cone}(P_{f(d)} \longrightarrow P_d).$$

The arrow is the incidence inclusion. Its restriction to U is an isomorphism onto $P_d|_U$, so C_d is the extension by zero of the projective at d in D. For $r \in R$ put

$$F_r = \mathrm{Cone}\big(P_{r^0} \oplus P_{r^1} \xrightarrow{(1,1)} P_{\widehat{r}}\big).$$

Thus each displayed cone is a projective complex in degrees $-1, 0$. There is a natural triangle

$$P_{r^0} \longrightarrow C_{\widehat{r}} \longrightarrow F_r \longrightarrow P_{r^0}[1]. \tag{19}$$

Index a family of objects by the vertices of P' as follows:

vertex	a^0	$\widehat{r}$	r^0	v^0	a^1	r^1	v^1
T_{vertex}	$P_{a^1}[1]$	$P_{r^1}[1]$	F_r	$P_{v^1}[1]$	C_{a^0}	$C_{\widehat{r}}$	C_{v^0}

$(v \in V)$.

We check its entire Hom algebra, including the new F_r terms. Let j_0 and i be the lower- and upper-supported embeddings. For the indicated d, e and $u, t \in U$, the projective resolutions above give

$$\begin{aligned} R\,\mathrm{Hom}(C_d, C_e) &= \mathrm{Hom}_D(P_d, P_e), \\ R\,\mathrm{Hom}(P_u[1], P_t[1]) &= \mathrm{Hom}_U(P_u, P_t), \\ R\,\mathrm{Hom}(P_u[1], C_d) &= 0, \\ R\,\mathrm{Hom}(C_d, P_u[1]) &= \mathrm{Hom}_U(P_{f(d)}, P_u). \end{aligned}$$

All complexes on the right are in degree zero. These comparisons include the usual incidence maps.

For the remaining terms, direct application of the same two-term resolution gives

$$R\,\mathrm{Hom}(C_d, P_{r^0}) = 0, \qquad R\,\mathrm{Hom}(C_{\widehat{r}}, C_d) \xrightarrow{\sim} R\,\mathrm{Hom}(P_{r^0}, C_d).$$

For $d \in V^0$, the first vanishing is the cancellation of two identical incidence maps; for $d \in A^0 \sqcup \widehat{R}$ both terms are zero. In the second identity both terms evaluate the same lower projective on the bottom of B. Applying (19) now yields

$$\begin{aligned} R\,\mathrm{Hom}(F_r, C_d) &= 0, & R\,\mathrm{Hom}(C_d, F_r) &\simeq \mathrm{Hom}_D(P_d, P_{\widehat{r}}), \\ R\,\mathrm{Hom}(F_r, F_s) &\simeq \mathrm{Hom}_R(P_r, P_s), & R\,\mathrm{Hom}(F_r, P_u[1]) &\simeq \mathrm{Hom}_U(P_{r^1}, P_u). \end{aligned}$$

In the other direction,

$$R\operatorname{Hom}(P_u[1], F_r) \simeq \begin{cases} k[0], & u = v^1,\ v \in V,\ r \le q_B(v), \\ 0, & \text{otherwise.} \end{cases}$$

Indeed $\operatorname{Hom}(P_u, P_{r^0})$ is exactly the missing-bottom mixed bimodule, and the two shifts in the triangle cancel. These identities prove, for every pair of target vertices,

$$R\operatorname{Hom}(T_x, T_y) = \begin{cases} k[0], & y \le x \text{ in } P', \\ 0, & \text{otherwise.} \end{cases}$$

Here is also a multiplication check, to exclude a twisted or zero-relation algebra. Let $a_* = \alpha(\min R)$; the least target vertex has image $P_{a_*^1}[1]$. Choose maps from every T_x to this object as follows: use incidence maps on the shifted upper projectives; on C_d use its degree -1 component $P_{f(d)} \to P_{a_*^1}$; on F_r use the P_{r^1} component. Whenever $y \le x$, composition with the chosen map from T_y induces an isomorphism

$$\operatorname{Hom}(T_x, T_y) \longrightarrow \operatorname{Hom}(T_x, P_{a_*^1}[1]).$$

For the families without F this follows from the displayed projective resolutions. For maps into F_r from a shifted P_{v^1}, the representative has components $(1, -1)$; its P_{r^1} component is a unit. For maps into F_r from C_d, that same component generates the quotient: homotopies affect only the P_{r^0} component. The maps between two F terms are the natural incidence maps of R. Maps out of F to an upper projective are induced by $C_{\widehat{r}}$, with this component again a unit. This exhausts the nonzero cases. All assertions use only cancellation of identity maps and coefficients $1, -1$. Normalize the generator a_{yx} by requiring its composite to the least object to equal the chosen map from T_x. Associativity then gives $a_{zy}a_{yx} = a_{zx}$ for $z \le y \le x$. Consequently $\operatorname{End}(T)^{\mathrm{op}} \cong kP'$ including multiplication.

The shifted upper projectives are summands. Their triangles with C_d recover all P_d for $d \in A^0 \sqcup \widehat{R} \sqcup V^0$. Triangle (19) recovers the remaining P_{r^0}. Hence T generates and is tilting. Let $\mathcal{T} : \mathcal{D}_{P'} \to \mathcal{D}_P$ be the resulting equivalence, sending the target projectives to this family.

Finally $Q_{P'}(P_r \boxtimes P_0) = P_{\alpha(r)^0}$ and $Q_{P'}(P_r \boxtimes P_1) = P_{\alpha(r)^1}$. Their images are respectively $Q_P(P_r \boxtimes P_1)[1]$ and $Q_P(P_r \boxtimes \operatorname{Cone}(P_1 \to P_0))$. To check that the algebra normalization preserves all these maps, write

$$X_r = P_{\alpha(r)^1}[1], \qquad Y_r = C_{\alpha(r)^0}.$$

Use the incidence maps on $X_s \to X_r$ and the componentwise incidence chain maps on $Y_s \to Y_r$ for $r \le s$. Let $\delta_r : Y_r \to X_r$ be the connecting map, represented by the identity on the degree -1 term of the cone. The mixed map $Y_s \to X_r$ is $X_s \to X_r$ composed with δ_s. It equals δ_r composed with $Y_s \to Y_r$ as a chain map.

Let $h_Z : Z \to P_{a_*^1}[1]$ be the maps used for normalization above. On X_r it is the incidence map; on Y_r it is the same incidence map on its degree -1 term. Hence for each of the three displayed types of base arrow $b : Z \to Z'$ one has $h_{Z'}b = h_Z$, with coefficient $+1$. These arrows already are the normalized incidence generators. No further rescaling of a base arrow or of an individual summand is needed. Moreover for any normalized arrow $a : W \to Z$, including an arrow involving F_r, one has $h_{Z'}ba = h_W$; uniqueness of the normalized generator shows that the mixed composite is also preserved. This checks the base arrows and their compositions with the full tilting algebra simultaneously. On C_2 the objects X_r, Y_r and the connecting map are exactly the images of the two

projectives and their arrow under $\mathbb{S}_{C_2}^{-1}[1]$. Thus the comparison is natural on the base projective category, and extends as an enhanced functor identity:

$$\mathcal{T} Q_{P'} \simeq Q_P(\mathrm{id} \boxtimes \mathbb{S}_{C_2}^{-1})[1].$$

Taking $G = \mathcal{T}^{-1}$ proves (18). □

B.1 Verification for pyramid blocks

We now verify Theorem 6.4, including the identification of the actual gluing relations.

Verification of the pyramid transfer. The interval $I(p_m)$ is the Boolean poset of compositions of m: a composition is represented by its concatenation of pyramid blocks, and a cover removes one cut between consecutive parts. For completeness, a legal downward move on a concatenation of pyramids replaces one pyramid by two consecutive pyramids. An inner split would leave the moved subpath followed by a zero, contrary to the movability condition. Iterating gives only such concatenations; conversely each removal of a cut is the legal full slide merging two consecutive pyramids. This describes all lower elements and covers. In particular, the index rule of Proposition 2.2 applies using just the numbers of current blocks, independently of their lengths.

Put $x \in I(p_a)$, $z \in I(p_{b-1})$ and $R = I(p_a) \times I(p_{b-1})$. On the source base $I(p_a p_b)$, mark the first internal cut in p_b; on the target base $I(p_{a+1} p_{b-1})$, mark the last internal cut in p_{a+1}. Writing 0 for an unremoved cut and 1 for a removed cut, both bases are $R \times C_2$. At coordinate 0 their lower word is exactly $x\,10\,z$. At coordinate 1, it is respectively $x\,\lambda(z)$ or $\rho(x)\,z$, where λ increases the first pyramid of z by one and ρ increases the last pyramid of x by one. Neither operation changes the number of blocks. The covers within coordinate 1 therefore identify its lower truncation with $L_{p_a p_{b-1}}$ on both sides.

Set

$$A = L_{p_a} \times I(p_{b-1}), \qquad B = I(p_a) \times W_{p_{b-1}}.$$

Here A has its bottom section and B its index-one bottom section, as required in Theorem B.1. The coordinate-zero part is $A^0 \sqcup \widehat{R} \sqcup B^0$: over (x, z) the three parts have indices $i \le b(x)$, $i = b(x) + 1$, and $i > b(x) + 1$. The coordinate-one part is $A^1 \sqcup B^1$, with its cut after $b(x)$.

We check the mixed relations, including the exceptional index. On the source, removing the marked cut merges block number $\ell = b(x)+2$. All of A^0 crosses with its index unchanged; $\widehat{R}$ crosses to the bottom of B^1. On B^0, local index $j > 1$ crosses to the same local index in B^1, whereas $j = 1$ is the forbidden index $i = \ell$. It must first increase to $j = 2$; if this point does not exist, no crossing from that fibre exists. Subsequent covers cannot decrease a positive local index greater than one to one. Thus the mixed relation on $B^0 \times B^1$ is exactly $b_0 \le b_1$ in B, with $b_1 \in B \setminus R$. Conversely every such relation is obtained by crossing directly for $j > 1$, or first increasing to two for $j = 1$, and then following the covers of B. All other mixed relations are exactly those in P above, by the marked boundary formula.

On the target, the removed cut merges block number $b(x) + 1$. Its forbidden point is now $\widehat{R}$, which can always first increase to the bottom of B^0. Consequently it still reaches the bottom of B^1. Every point of B^0 crosses directly with its local index unchanged. The mixed relation on $B^0 \times B^1$ is therefore the full comparison in B. The other relations are unchanged. This identifies the actual lower truncations with P and P' of Theorem B.1, not just their fibre counts.

That theorem constructs the relative equivalence, with base correction $\mathrm{id}_{\mathcal{D}_R} \boxtimes \mathbb{S}_{C_2}[-1]$. The generalized relative extension criterion (Proposition 5.3) transports its actual mixed bimodule and multiplication to arbitrary exterior strings. □

C The canonical shape cut and its limitations

C.1 The singleton-cut tilting algebra

Put $P = J(w)$, let D be its lower-shape part and U its top-shape part, and identify $U = I(w)$. Then D is a lower ideal and U an upper filter. The map $f : D \to U$, $f(e_i(x)) = x$, is monotone, and

$$d \le u \text{ in } P \quad \Longleftrightarrow \quad f(d) \le u \text{ in } U. \tag{20}$$

This follows also from [1, Proposition 5.6]: the top of the chain is the least top-shape point above d. Thus the mixed bimodule has values

$$M(u,d) = k \quad \text{if } f(d) \le u, \qquad M(u,d) = 0 \text{ otherwise,}$$

with all nonzero incidence compositions equal to the corresponding basis map. In the order D, U, the algebra is $\left(\begin{smallmatrix} kD & 0 \\ M & kU \end{smallmatrix}\right)$.

The underlying equivalence is the singleton flip-flop of [2, Corollary 1.3]. In that result take $X = D$, $Y = U$, and the order-preserving map $f : D \to U$: its positive and negative mixed relations are exactly the two relations below. We retain the explicit tilting calculation to specify its object and multiplication in our covariant convention. For any finite posets D, U and monotone $f : D \to U$, let P_f retain the two internal orders and have mixed relations $d \le u$ exactly when $f(d) \le u$. Let P_f^{fl} instead have mixed relations $u \le d$ exactly when $u \le f(d)$. These rules define posets.

Proposition C.1 (Explicit singleton-cut tilting)**.** *For every field k, write $P_z = k\,\mathrm{Hom}_{P_f}(z, -)$. For $d \in D$ put*

$$C_d = \mathrm{coker}(P_{f(d)} \longrightarrow P_d), \qquad T = \bigoplus_{u \in U} P_u \;\oplus\; \bigoplus_{d \in D} C_d[-1].$$

Then T is a tilting object and

$$\mathrm{End}_{D^b(\mathrm{Rep}_k P_f)}(T)^{\mathrm{op}} \cong kP_f^{\mathrm{fl}}.$$

Consequently the two bounded derived representation categories are equivalent.

Proof. The indicated map is the identity on the upper part of P_d and zero on its lower part. Equation (20), with P_f in place of P, gives an exact sequence

$$0 \longrightarrow P_{f(d)} \longrightarrow P_d \longrightarrow C_d \longrightarrow 0. \tag{21}$$

The restriction of C_d to D is the representable at d, and its restriction to U is zero. This is a length-one projective resolution in the full category, not just in a subcategory.

By Yoneda and this resolution, all the derived Hom groups between the summands of T are concentrated in degree zero. Their degree-zero values are

$$\begin{aligned} \mathrm{Hom}(P_u, P_v) = k \quad &\Longleftrightarrow \quad v \le u, \\ \mathrm{Hom}(C_d[-1], C_e[-1]) = k \quad &\Longleftrightarrow \quad e \le d, \\ \mathrm{Hom}(C_d[-1], P_u) = \mathrm{Ext}^1(C_d, P_u) = k \quad &\Longleftrightarrow \quad u \le f(d), \\ \mathrm{Hom}(P_u, C_d[-1]) = 0. & \end{aligned}$$

Unlisted groups are zero. In detail, applying $\mathrm{Hom}(-, P_u)$ to (21) gives zero in degree zero from P_d and $P_u(f(d))$ in degree one from $P_{f(d)}$. Applying $\mathrm{Hom}(-, C_e)$ gives $C_e(d)$ in degree zero and zero in degree one. Finally P_u is projective and $C_d(u) = 0$. These computations also exclude negative

degrees after the stated shifts. The triangles of (21), together with the P_u, recover all P_d from T. Since a finite poset algebra has finite global dimension, its projectives generate its bounded derived category. This proves generation.

It remains to verify multiplication. Use the canonical representable maps as bases inside U and inside D. For the mixed group use the class of $P_{f(d)} \to P_u$ in the Hom complex of (21). If $e \le d$, the canonical map $C_d \to C_e$ lifts to the diagram of resolutions with terms $P_d \to P_e$ and $P_{f(d)} \to P_{f(e)}$. The diagram commutes with coefficient 1, because $f(e) \le f(d)$ and all maps are representable maps. Composition with this diagram and composition with $P_u \to P_v$ therefore send a mixed basis class to the corresponding basis class, with coefficient 1 whenever the endpoints are comparable. There are no composable pairs of mixed arrows. These checks exhaust the products; identity maps are preserved. Taking the opposite algebra gives exactly the incidence basis and multiplication of P_f^{fl}.

The tilting argument of Lemma 2.6 gives the equivalence. All entries in the resolutions and product comparisons are 0 or 1, so no restriction on the field is used. □

C.2 Why its target is not the exchanged interval

Applying Proposition C.1 to the shape cut constructs a specific algebra, kP_f^{fl}. Identifying it with $kJ(AvuB)$ is an additional assertion and can fail for this choice of cut.

Proposition C.2. *For $w = (10)(1100)$ the flipped shape poset P_f^{fl} has two maximal elements. In particular it is not isomorphic to $J((1100)(10))$, and its incidence algebra is not isomorphic to that target incidence algebra over any field.*

Proof. The lower interval has just $x_0 = 101010 < x_1 = 101100$. Its sole cover merges at position $\ell = 3$. The lower-shape part consists of chains

$$a_0 < a_1 < a_2 < a_3, \qquad b_0 < b_1 < b_2,$$

where $a_i = e_i(x_0)$ and $b_i = e_i(x_1)$, with additional generating relations $a_i < b_i$ for $i = 0, 1, 2$. This follows directly from Proposition 2.2. It has exactly two maximal points, a_3, b_2. In the flipped poset all upper-shape points lie below lower-shape points, so these remain exactly its maximal points. Every $J(z)$ has a unique maximum. Moreover, for covariant representations, the simple projective modules of a poset algebra are exactly those at maximal points: the representable at z is simple precisely when nothing is strictly above z. Thus the two algebras have different numbers of simple projectives, an isomorphism invariant over every field. □

This is a failure of the full shape-cut candidate, not a counterexample to Problem 1.1. The two nine-element intervals themselves are known to be derived equivalent by the smaller flip-flop described in [1, Section 9]. Nor does Proposition C.2 rule out an additional tilting step from the displayed flipped algebra.

D Obstructions for restricted base corrections

D.1 A strict comparison

The example in this appendix involves 10 and 1100, not the pair in Corollary 5.5.

Proposition D.1 (A necessary base twist)**.** *Let $w = (10)(1100)$ and $w' = (1100)(10)$, and identify their lower intervals S by the product permutation. There is no k-linear triangle equivalence G :*

$\mathcal{D}_{L_w} \to \mathcal{D}_{L_{w'}}$ *with* $GQ_{L_w} \simeq Q_{L_{w'}}$. *Thus setting* $K = \mathrm{id}$ *in Proposition 5.3 is in general too restrictive.*

Proof. Both S are the two-point chain $x_0 < x_1$, with $x_0 = 101010$ and $x_1 = w$ or w'. Order the seven vertices of each L as $e_0(x_0), \dots, e_3(x_0), e_0(x_1), \dots, e_2(x_1)$. Proposition 2.2 gives their zeta matrices

$$C = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad C' = C + E_{4,7},$$

where $E_{4,7}$ is a matrix unit. The two pulled-back base projectives have the same dimension-vector matrix on both sides:

$$Q = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

For a zeta matrix C, the Euler form in the simple basis is $E = C^{-1}$, and the Serre action is $\Sigma = E^{-1}E^t = C(C^{-1})^t$. The latter identity follows from $\chi(a,b) = \chi(b, \Sigma a)$. Therefore the matrix of $\chi(QP_i, \mathbb{S}_L^4 QP_j)$ is $Q^t E \Sigma^4 Q$. Direct multiplication of the displayed integer matrices yields

$$Q^t C^{-1}\big(C(C^{-1})^t\big)^4 Q = \begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}, \qquad Q^t (C')^{-1}\big(C'((C')^{-1})^t\big)^4 Q = \begin{pmatrix} -1 & -1 \\ 0 & 0 \end{pmatrix}.$$

An equivalence commutes with Serre functors and preserves the Euler form. The proposed strict relative identification would therefore identify these matrices entry by entry, a contradiction. All entries are Euler characteristics over k, so this argument works in every characteristic; the integer invariant is not reduced modulo the characteristic. □

Remark D.2. This obstruction does not contradict Theorem 3.2 or rule out the extension criterion with a nontrivial K. For example, in the projective basis of the two-point base chain, its Serre action is $K_0 = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The two fourth-moment matrices above satisfy $M = K_0^t M' K_0$. Thus this particular obstruction disappears when the base Serre functor is allowed. This equality is a necessary numerical check, not a construction of G or of its natural compatibility. Theorem 6.1 constructs (G, K) when one factor is 10. For arbitrary factors a uniform full-context equivalence is impossible by Theorem 4.2.

D.2 Products of Serre powers

The elementary-block construction does not extend merely by choosing other integer powers of the two base Serre functors. The following obstruction concerns the sufficient relative criterion, not the original block-permutation problem.

Proposition D.3. *Let $u = 1100$, $v = 111000$, and identify the bases of L_{uv} and L_{vu} with $S = S_u \times S_v$. There is no enhanced equivalence $G : \mathcal{D}_{L_{uv}} \to \mathcal{D}_{L_{vu}}$ satisfying*

$$GQ_{L_{uv}} \simeq Q_{L_{vu}}(\mathbb{S}^p_{S_u} \boxtimes \mathbb{S}^q_{S_v})[h]$$

for any integers p, q, h and any field k.

Proof. For $X = L_{uv}$ or L_{vu}, let C_X be its zeta matrix in a linear extension and let Q_X be the dimension-vector matrix of the pulled-back base projectives. Set

$$M_n^X = Q_X^t C_X^{-1}\big(C_X (C_X^{-1})^t\big)^n Q_X.$$

If the proposed relative equivalence exists and its base action on the projective Grothendieck basis is K, preservation of Euler pairings and commutation with Serre give the necessary identities

$$M_n^{L_{uv}} = K^t M_n^{L_{vu}} K \qquad (n \geq 0). \tag{22}$$

These are integer Euler characteristics, including in positive characteristic.

Here S_u is the chain on $1010, 1100$, and S_v is the diamond with ordered vertices

$$101010, \quad 101100, \quad 110010, \quad 111000.$$

In those projective bases their Serre matrices are

$$U = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & -1 & 0 \\ -1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

They satisfy $U^3 = -I_2$ and $V^3 = I_4$. Consequently the congruence in (22) for $K = (-1)^h U^p \otimes V^q$ depends only on p, q modulo 3.

Here is an explicit finite certificate for all nine residues. Order the base pairs (x, y) lexicographically in the displayed orders. For each pair use the vertices $e_i(xy)$ on the source and $e_i(yx)$ on the target, with $0 \leq i \leq b(x) + b(y)$. This gives linear extensions with 36 vertices on both sides. In these coordinates Q_X has entry 1 exactly when its column base pair is below its row base pair. The zeta matrices are obtained from the cover rule of Proposition 2.2; thus no choice of mixed maps enters this computation. Matrix multiplication gives the following violations of (22). Entry indices are one-based; $M_n = M_n^{L_{uv}}$ and $M'_n = M_n^{L_{vu}}$.

p	q	n	(i, j)	$(M_n)_{ij}$	$(K^t M'_n K)_{ij}$
0	0	4	$(1, 4)$	-1	0
0	1	4	$(1, 4)$	-1	0
0	2	4	$(1, 4)$	-1	0
1	0	4	$(1, 4)$	-1	0
1	1	4	$(1, 4)$	-1	0
1	2	4	$(1, 4)$	-1	0
2	0	4	$(1, 1)$	0	-1
2	1	5	$(1, 5)$	0	-1
2	2	4	$(1, 4)$	-1	0

Every residue violates a necessary identity. Periodicity and the irrelevance of the overall sign prove the assertion for all integer powers and shifts. □

Remark D.4. The proposition excludes this entire factorwise Serre family, not arbitrary base autoequivalences. More generally, (22) implies simultaneous similarity of $A_n = (M_0)^{-1}M_n$: namely $A'_n = KA_nK^{-1}$. The matrix M_0 is invertible because the pullback is fully faithful and the base Euler form is nonsingular. Therefore traces of all words in the A_n are necessary invariants independent of K. Equality of such traces does not identify the corresponding boundary functors and cannot replace their natural comparison. No counterexample to Problem 1.1 follows here.

Remark D.5 (The refined correction is realized)**.** For $(u, v) = (1100, 111000)$ the base is C_2^3. Label its coordinates (a, b, c) with the last two belonging to S_v. The marked coordinate in Theorem 6.4 is b; under the original target factor ordering its relabeling is $\Pi : (a, b, c) \mapsto (c, a, b)$. Thus that theorem constructs an actual relative equivalence with base correction

$$\Pi \circ (\mathrm{id} \boxtimes \mathbb{S}_{C_2} \boxtimes \mathrm{id})[-1].$$

Its matrix is $-P(I_2 \otimes U \otimes I_2)$, where P denotes the coordinate permutation. All the moment identities (22) now follow from the natural relative comparison. The construction mixes the internal factors of the original base, so Proposition D.3 does not apply to it. An overall shift by two changes $[-1]$ to $[1]$ without changing that matrix.

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